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RESEARCH ARTICLE

Fine-Grained Relation Extraction for Drug Instructions Using Contrastive Entity Enhancement

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ABSTRACT The extraction of relations between drug-related entities from drug instructions is essential for clinical diagnostic decision-making and drug use regulations, which is a critical task. However, due to the complexity of the textual descriptions in drug instructions, it is challenging to extract fine-grained relations, even with a considerable amount of training data. Moreover, since manually-labeled, high-quality datasets in the pharmaceutical domain are typically expensive, obtaining an extensive and accurate training dataset could be challenging. To overcome the above challenges, this paper proposes a drug relation extraction framework that combines entity information enhancement and contrastive feature learning, which can better extract fine-grained relations with limited data. More specifically, a sample generator creates a group of different samples with role semantic information from the training set, an entity encoder embeds the entity role information and context information to enhance the semantic representation, and a contrastive learning module employs a hybrid loss function to learn inter-sample and intra-sample differences. Empirical study indicates that the contrastive entity enhancement approach can achieve higher extraction accuracy and has better generalization capability. More specifically, the experimental results show that the F1 value of the model can reach 0.8892, which provides a 7.13% improvement compared to the baseline pre-training method.

INDEX TERMS Contrastive learning, drug instruction, entity enhancement, fine-grained, relation extraction.

1. INTRODUCTION

Drug instructions contain a wealth of medical knowledge. Mining entities and relations automatically from drug instructions is a challenging but crucial task in building a medical domain knowledge base and developing knowledge-based

healthcare systems. There are two main research strands in relation extraction: 1) pipeline methods [1], which first identify entities and then categorize the entity pairs with relation labels, and 2) joint extraction methods [2], [3], [4] that extract entities and relations together. Joint methods have recently gained more attention since pipeline methods suffer from error propagation. However, a recent comparative study [1] suggests pipeline methods still have great potential when

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carefully managing error propagation. Another study [5] also demonstrates that by using entity pair calibration, we can effectively eliminate noise in entity pairs and extract accurate relations from drug instructions. In addition, we also found that the joint extraction model requires a large amount of data to achieve model convergence. In this paper, we continue our research on pipeline relation extraction, focusing on a critical problem in drug information extraction: making the most out of the expensive training dataset labeled by domain experts.

Pharmaceutical data annotation requires expert knowledge; sometimes, experts need to discuss and get a consensus on a specific label. As a result, the amount of manually labeled, high-quality data is often limited, making it challenging to generate good results. Several approaches have been made to address such a problem by enhancing the generalization capability of neural models, such as constructing a deeper model structure [6], adding regularization to the loss function [7], multi-feature fusion learning [8], data enhancement [9], dropout [10], etc. In this paper, we mainly focus on data enhancement and contrastive learning.

The metadata of entities (e.g., types, roles) can enhance the data semantics [11]. For example, encoding the information of entity types into neural networks can improve the performance of relation extraction [12], [13]. The basic idea is while extracting certain relations. We can use the relation domain and range types to emphasize correct entity pairs. However, such approaches have two limitations: 1) it relies on a precise and predefined taxonomy, which may not always be available, and 2) it cannot deal with the noise caused by entities in the same entity type but different roles (as depicted in Table 1).

Taking a closer look at example #1 in Table 1, the bolded fields represent entities. Still, not all entities are related to each other, e.g., the primary entity in the “should not be used together with” relation in this sentence must be “paracetamol.” If the model learns the role semantics of these entities, i.e., which entities are primary (the main drug described in the instruction) or secondary (other entities mentioned in the education) in the drug instruction sentence, it can reduce many meaningless combinations of entity pairs.

TABLE 1. Sample drug instruction text with multiple entities and relations (entities in bold).

Number	Sentence
1	Compound paracetamol should not be used together with chloramphenicol , barbiturates (such as phenobarbital), etc.
2	Amikacin sulfate injection should not be compatible with amphotericin B , cephalon thiophene , sulfadiazine , and tetracycline injections , etc.

In addition to the above “should not be used together with” relation, we can see that “barbiturates” is a superclass of “phenobarbital,” “chloramphenicol,” and “barbiturates”

have no obvious interaction. There is no obvious interaction between “chloramphenicol” and “barbiturates.” These differences require the model to learn more fine-grained features in the sentence. Contrastive learning can be used to grasp minor differences by aligning positive samples and amplifying the differences between negative ones. However, existing contrastive learning methods focus on producing different representations of the same sample without considering the different relations between different samples, as in [12]. Moreover, syntactic or grammatical features like the length of the sentence are not considered during positive and negative sample construction. Both could lead to learning insufficient distinctive features. Data augmentation can be used for the comparison of samples [13]. Still, it risks changing the semantics of the sentence, especially in certain languages like Chinese, in which altering basic language units may significantly impact the meaning.

To address the above problems, we propose a novel Contrastive Entity Enhancement (CEE) model that integrates both entity role information and contextual information to enable the model to learn the semantic roles of entities better. We also utilize an intra-sample and inter-sample contrastive learning network to capture subtle differences within this semantic information. Additionally, during the sample construction process, we consider the positive impact of sentence feature changes on the model. The main contribution of this paper is two-fold:

1) we propose an entity enhancement encoding method based on BERT (wwm). This method integrates entity role information as well as context information so that we can better identify drug-related entities as well as their roles in the sentences, ultimately reducing noise in relation extraction;

2) we propose an intra-sample and inter-sample contrastive learning network to differentiate the enhanced entity vectors within the same entity pair and across different entity pairs.

Empirical results indicate that our model can accurately extract drug knowledge even with limited labeled data. The remainder of this paper is organized as follows: Section II discusses the related work; Section III presents the CEE model focusing on the entity enhancement and contrastive learning framework; Section IV evaluates our approach with realistic datasets before Section V concludes.

II. RELATED WORKS

This section reviews the related work from the following three aspects. The first presents the recent advances in pre-training models and entity enhancement. The second introduces the latest information about contrastive learning. Finally, we researched the current situation of knowledge extraction in the medical and pharmaceutical domains.

A. PRE-TRAINING MODELS AND ENTITY ENHANCEMENT

After traditional machine learning [14], [15] methods, neural networks [16] have been widely used in knowledge extraction. With massive training data, the pre-trained model is very useful for many NLP tasks. ELMo [17] presents a

context-dependent text representation method based on bidirectional LSTM, which can effectively learn semantic features and achieve good results on multiple tasks. With the huge performance improvement brought by the attention mechanism, Vaswani et al. [18] proposed the Transformer model that outperforms traditional neural network structures such as CNN and RNN. The BERT [19] (Bidirectional Encoder Representation from Transformers) is developed based on the transformer, and it is proven efficient by two tasks: Mask Language Model (MLM) and Next Sentence Prediction (NSP).

Several efforts have been made to improve BERT. Wu and He [20] combined the pre-trained model and entity information in the text for relation extraction, which effectively retrieves high-quality results, proving the importance of entity information in relation extraction. In [20], two entity vectors together with sentence representation vectors encoded by BERT are concatenated, but the utilization of rich entity information needs to be thoroughly discussed. Wan et al. [21] combines information about two entities and entity context for relation classification and performs well in relation classification tasks. Wang et al. [22] uses the marker embedding of the position of the entity pair as a contextual relation to representing the SOTA obtained in the zero-shot scenario. Chen et al. [23] splices all tags at the beginning of the sentence to represent the final sentence vector and achieves good results. Meanwhile, the BERT model is originally proposed for English Corpus. Using the same “MASK” mechanism for the Chinese Corpus will result in information loss. BERT (wwm) [24] was proposed to address this issue by masking the whole Chinese word. BERT (wwm) showed better experiment results on the data set CMRC 2018. Zhang et al. [25] used BERT (wwm) for relation extraction and achieved better results than BERT for Chinese text processing. In this paper, we also use BERT (wwm) as the pre-trained model and emphasize enhancing the entity information during encoding.

B. CONTRASTIVE LEARNING

Contrastive learning was initially applied in the field of CV. It uses image rotation, color change, and cropping transformation to generate samples slightly different from the original image. These altered images are treated as positive samples of the original image, and the goal of the network is to minimize the distance of positive samples in the feature representation vector space [26], [27], [28]. Later, contrastive learning was used in NLP. Wang et al. [29] proposed using a pre-trained language model for contrastive learning, slightly perturbing the sentence semantics by sampling, and using an unsupervised method to construct negative samples in large data sets. Khosla et al. [30] first extended the self-supervised contrastive learning method to the supervised settings, pulling sample clusters belonging to the same category together in the embedded space and pushing away sample clusters from different categories. Gunel et al. [31] added the loss

of contrastive learning in the fine-tuning stage of the BERT, extending supervised contrastive learning to the classification tasks. Miao et al. [32] used a combination of adversarial training and contrastive learning to increase the difficulty of contrastive sample learning and achieved good results. Gao et al. [10] proposed the SimCSE model with supervised and unsupervised training methods. Using the dropout in the pre-training model as a means of data enhancement has a better effect than previous methods. Zhang [13] proposed an S-SimCSE that achieves the SOTA results with different dropout rates based on SimCSE, showing that the effect of having an adaptive dropout rate is important for the network. Wu et al. [33] proposed the ESimCSE model, which is mainly based on the SimCSE model by optimizing the positive and negative samples through whole word repetition and momentum comparison, thereby making certain improvements. Chi et al. [34] demonstrated its effectiveness with fair representation by mixing representations of examples with different task labels based on supervised contrastive learning. In this paper, we investigate ways to construct contrastive samples for different relations within and across samples while considering using syntactic features such as text length to create an unbiased and diverse training dataset.

C. KNOWLEDGE EXTRACTION IN THE MEDICAL AND PHARMACEUTICAL DOMAIN

Knowledge extraction in the medical field has become a research hotspot in recent years. Texts in the medical field have distinct characteristics, such as the multiple presences of entity types, entity names, and complex relations in a sentence. Gong et al. [35] used a drug entity relation extraction method combining GRU and CNN, mainly extracting adverse drug reactions (ADRS) between drugs, and achieved an overall accuracy of 75% in DDIE-xtraction2013. Huang et al. [36] proposed that EGFI uses a comprehensive pre-trained language model BioBERT on a biomedical corpus to extract and consolidate drug interactions from large-scale medical literature text data and demonstrated that EGFI could be used to identify new drug relations. Still, it focused only on drug-drug interactions and cannot cater to other drug-related relation extraction. Yuan et al. [37] proposed a biomedical pre-trained language model (KeBioLM) that effectively analyzes medical text. Despite many efforts, mining sophisticated entities and relations from medical corpora remains a challenging task.

III. CEE MODEL

In this section, we first elaborated on the sample generator, which creates a group of samples with role semantic information. Then, we interpret how to embed entity and role information in the encoder particularly. Finally, we detail the nuances of capturing models through contrastive learning.

The CEE model mainly consists of three parts: the sample generator (G), entity encoder (E), and feature contrastive module (F):

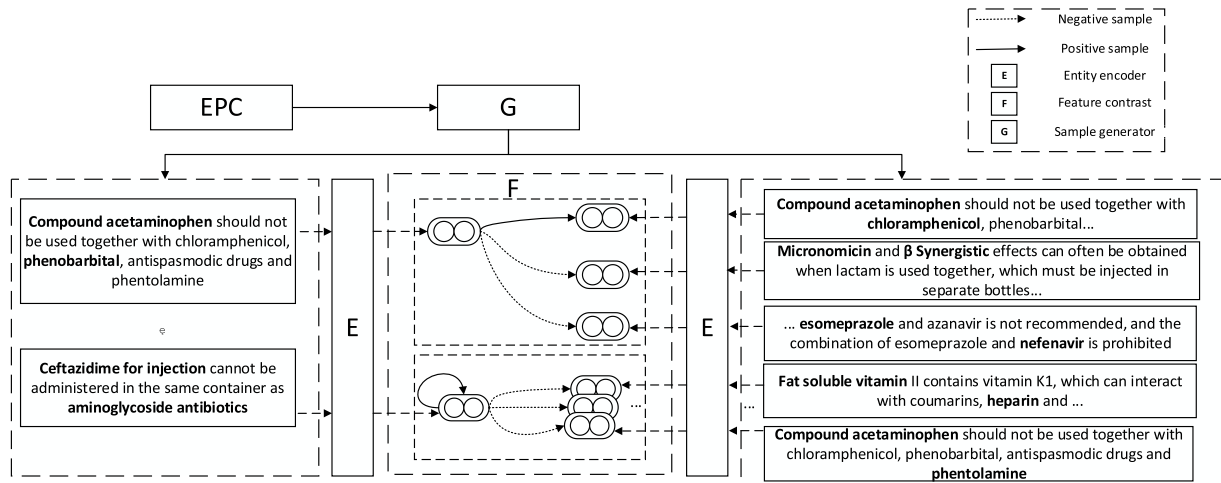


FIGURE 1. The overall architecture of the CEE model.

- 1) The sample generator (G) first leverages an Entity Pair Calibration (EPC) module (described in [5]) to filter out irrelevant entity pairs. Then, it investigates the features of the sentences in the dataset and prepares the positive and negative samples without changing the semantics;
- 2) The entity encoder (E) uses a modified BERT (wwm) encoding schema to incorporate both entity role and context information into the sentence representation;
- 3) The feature contrastive learning module (F) further examines different features inter-sample and inter-sample to improve the generalization ability of the model. As shown in Figure 1.

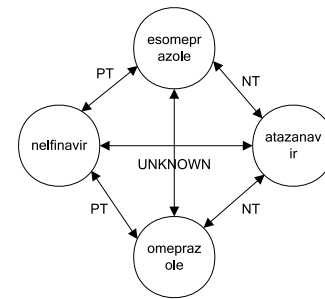


FIGURE 2. Multiple relations in a sentence, NT = not recommended to be used together, PT = prohibit use together.

A. SAMPLE GENERATOR

In this subsection, we first elaborated on the specific composition of the sample generator and then introduced the impact of changing the characteristics of the same sentence on the model.

Fine-grained relations often occur in drug instruction, with many entities often appearing side-by-side in the same sentence in a drug instruction. Not every entity pair has a relation between these entities, and there may be completely different relations between different entity pairs.

As shown in Figure 2. The relation between “esomeprazole” and “atazanavir” is “NT,” and the relation between “esomeprazole” and “nelfinavir” is “PT.” To learn such fine-grained relations within a limited training set, the model must share features learned from different entity pairs and examine the differences.

The sample generator creates positive and negative samples in the following steps. A sample s in this paper is defined as $s := (S, e_1, e_2, r)$, where S is a sentence in a drug instruction, e_1 and e_2 are two drug-related entities in S , and r is the relation label for e_1 and e_2 in S . The sample generator first uses the entity pair calibration (EPC) module to verify the primary entity e_1 (the primary drug described in the drug manual) and the second entity e_2 (other drugs described in the sentence) [5]. Then, the sample generator

divides a sentence into multiple samples according to the number of entity pairs, and the samples separated by the same sentence have the same sentence information S and header entity e_1 . To capture the difference of intra-sample, the augmented samples $s^+ := (S, e_1, e_2, r)$, of s are obtained by data augmentation, s^+ is regarded as a positive sample of s , and the other samples are regarded as negative samples. Meanwhile, to capture the difference of inter-samples, two samples s_1 and s_2 are considered positive samples w.r.t. r if $s_1 \neq s_2 \wedge s_1 \cdot r = s_2 \cdot r$, else they are considered negative samples w.r.t. $s_1 \cdot r_1$ or $s_2 \cdot r_2$. Given a predefined sample batch B , B^+ is the augmented batch of B . The sample generator creates a matrix $M = \{B \cup B^+\} \times \{B \cup B^+\}$, where each entry in M represents a sample pair, and their polarity, i.e., if they form a positive or negative sample pair.

It is also observable that since a sentence could have more than two entities, it is possible to generate multiple samples from the same sentence. Many of such samples are positive sample pairs (the same relation applies to a series of drugs described in the sentence). This is beneficial for learning the complicated entities and relations within the sentence. However, this will create a bias in the training set. When BERT (wwm) is used to encode the sentence, syntactic or syntactic features such as length information and entity location information are embedded, and such information is identical for

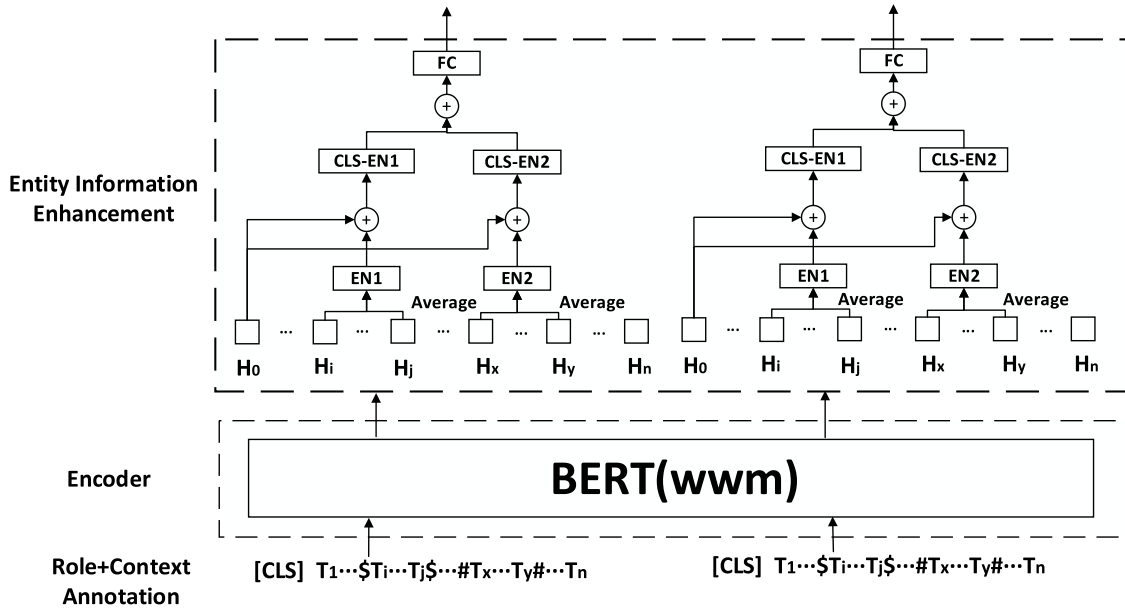


FIGURE 3. Detailed structure of the encoder.

TABLE 2. Examples of different methods for changing sentence features.

Features	Sentence
original	Paracetamol should not be used in combination with chloramphenicol, phenobarbital and antispasmodics.
change entity position	Paracetamol should not be used in combination with phenobarbital, antispasmodics and chloramphenicol.
change sentence length	Paracetamol should not be combined with chloramphenicol, phenobarbital and antispasmodics.

the positive samples from the same sentence. This could trick the model into believing this feature information entails positive samples. While this may be true on a local distribution (especially when the batch size or training set is small), it is not true on a global distribution. We intentionally change the characteristics of samples generated from the same sentence, as shown in Table 2.

B. ENHANCED ENTITY ENCODER

When generated, samples are fed to the encoder to get a semantic representation. Given the text characteristics of drug instructions, we chose BERT (wwm) to encode the text. There are two special positions in the BERT (wwm) model [24], namely “[CLS]” and “[SEP],” “[CLS]” is located at the head of the sequence to be used as the semantic representation of the whole sentence for text classification tasks. “[SEP]” is located at the end of the sequence to separate two different sentences.

Before fine-tuning, two special tags are inserted into the two entities for relation extraction in the text. Given an input

sentence S with primary and multiple secondary entities of n words $\{T_1, T_2, \dots, T_n\}$, to enable the BERT (wwm) model to capture the information of the entity, two special position marks “\$” and “#” are inserted at the beginning and end of the primary and secondary entities, respectively. Creating two entity sets, E_p and E_s . Then, the entity pairs can be produced by $E_p \times E_s$. On the one hand, these marks can locate the position of the entity in the sentence, and on the other hand, they can also strengthen the information of the entity.

The details of the encoder part are shown in Figure 3. For a given sentence containing two entities, the vector $\mathbf{H}$ of the hidden layer is encoded by BERT (wwm). $\mathbf{H}_i$ to $\mathbf{H}_j$ is the vector of the hidden layer encoded by BERT (wwm) on the entity, and then this vector is taken out and averaged.

$$\mathbf{H}_{\text{avg1}} = \frac{1}{j - i + 1} \sum_{t=i}^j \mathbf{H}_t \quad (1)$$

$$\mathbf{H}_{\text{avg2}} = \frac{1}{y - x + 1} \sum_{k=x}^y \mathbf{H}_k \quad (2)$$

Then add ($\cdot$) the vector $\mathbf{H}_0$ corresponding to “[CLS].” In this way, the final trained hidden layer combines the head vector $\mathbf{H}_0$ encoded by BERT (wwm) and the vector $\mathbf{H}_{\text{avg}}$ of the entity itself and position. Using the information of the entity itself and its location, the vector output by BERT (wwm) encoded is strengthened.

$$\mathbf{H}_1 = \mathbf{H}_0 + \mathbf{H}_{\text{avg1}} \quad (3)$$

$$\mathbf{H}_2 = \mathbf{H}_0 + \mathbf{H}_{\text{avg2}} \quad (4)$$

The enhanced vectors $\mathbf{H}_1$ and $\mathbf{H}_2$ of the entity information are obtained, and then the two enhanced vectors are summed up again by add ($\cdot$) to obtain the final fine-tuned vector $\mathbf{H}_{\text{all}}$.

And $\mathbf{H}_{all}$ is directly fed into the fully connected layer for fine-tuning, and finally, the relation between the entity pairs is output through Softmax ($\cdot$).

$$\mathbf{H}_{all} = \mathbf{H}_1 + \mathbf{H}_2 \quad (5)$$

$$\mathbf{H}_f = \mathbf{W}_1[\tanh(\mathbf{H}_{all})] + \mathbf{b}_1 \quad (6)$$

$$p = \text{Softmax}(\mathbf{H}_f) \quad (7)$$

where $\mathbf{W}_1$ is a weight, $\mathbf{b}_1$ is bias. p is the probability corresponding to the final label. Unlike Wu and He [20], we join the sentence representation vector to the two entities. The two vectors ($\mathbf{H}_1$ and $\mathbf{H}_2$) are added instead of concatenated. This results in faster computation while achieving better performance. Finally, the head vector $\mathbf{H}_0$ of the text encoded by BERT (wwm) and the vector $\mathbf{H}_{avg}$ of the entity itself and position are fully utilized so that the vector output by BERT (wwm) is strengthened and better experimental results can be obtained.

C. FEATURE CONTRASTIVE LEARNING

We enter the downstream fine-tuning stage when the sentence vectors are derived from the enhanced entity encoder. Using feature contrastive learning, we can combine the different features from different relations in different samples. For a given sample a: “esomeprazole and nelfinavir should be banned from being used together,” $\mathbf{h}_a$ is obtained through the entity enhancement encoder, and they should be closer together in the embedding space than the other samples labeled as “prohibit use together.” By inter-sample contrastive learning, the distance between samples with the same label can be minimized and the distance between samples with different labels maximized. In addition, the augmented samples $\mathbf{h}_a^+$ generated by dropout have the same semantics as A, and they should be closer together in the embedding space relative to the other semantics samples. The distance between samples with the same semantics can be minimized through intra-sample contrastive learning. By combining inter-sample and intra-sample contrastive learning, the model can learn to distinguish between fine-grained relations and achieve better generalization capability. To do that, we designed a features contrastive loss model, as illustrated in Figure 4.

To learn the difference between intra-sample, we should encourage the similarity between different views of the same sample but no other samples. We follow the method of SimCSE [10]. Given a batch of training data $\{x_i, y_i\}_{i=1}^n$, where x_i is sentence and y_i is the corresponding label, we use dropout as the data augmentation method, pass the same sample to BERT twice, and obtain the vector representation $\mathbf{h}_i$ of the original samples and the vector representation $\mathbf{h}_i^+$ of the augmented samples, where $\mathbf{h}_i$ and $\mathbf{h}_i^+$ have the same label y_i . Defining $\mathbf{N} = \{\mathbf{h}_i\}_{i=1}^{i=n} \cup \{\mathbf{h}_i^+\}_{i=1}^{i=n}$, the intra-sample contrastive loss is defined as:

$$\mathcal{L}_{ICL} = -\log \frac{e^{\text{sim}(\mathbf{h}_i, \mathbf{h}_i^+)/\tau}}{\sum_{\mathbf{h}_p \in \mathbf{N}} (e^{\text{sim}(\mathbf{h}_i, \mathbf{h}_p)/\tau})} \quad (8)$$

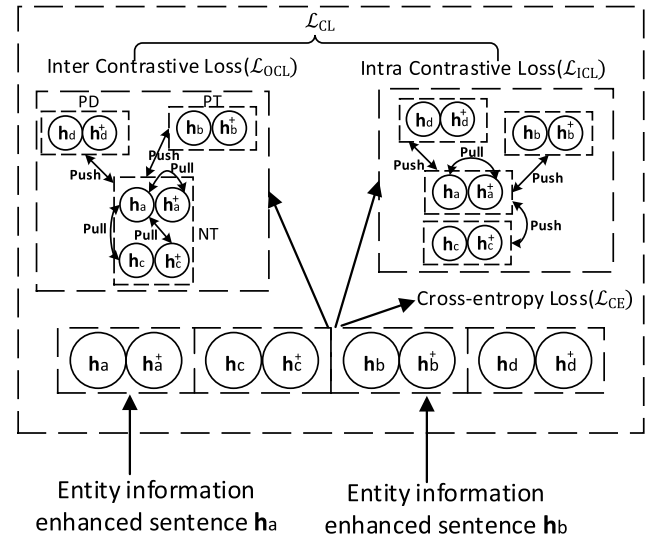


FIGURE 4. The total loss of CEE.

where τ is the temperature constant, $\text{sim}(\mathbf{h}_1, \mathbf{h}_2)$ is the cosine similarity $\frac{\mathbf{h}_1^T \mathbf{h}_2}{\|\mathbf{h}_1\| \cdot \|\mathbf{h}_2\|}$.

Meanwhile, samples given the same task label should be encouraged to learn the difference between inter-sample to minimize the similarity in the embedding space. This is consistent with the approach of Khosla et al. [30] for supervised contrastive learning. Formally, given a batch of augmented samples $\mathbf{N}$, where the second half is the augment view of the first half, and the positive sample set is defined as $\mathbf{P} = \{\mathbf{h}_p \in \mathbf{N}, (y_p = y_i) \wedge (p \neq i)\}_{i=1}^{i=p}$, the negative sample set is defined as $\mathbf{K} = \{\mathbf{h}_k \in \mathbf{N}, y_k \neq y_i\}_{i=1}^{i=k}$, the inter-sample contrastive loss is defined as:

$$\mathcal{L}_{OCL} = -\frac{1}{|\mathbf{P}|} \sum_{\mathbf{h}_p \in \mathbf{P}} \log \frac{e^{\text{sim}(\mathbf{h}_i, \mathbf{h}_p)/\tau}}{\sum_{\mathbf{h}_p \in \mathbf{P}} (e^{\text{sim}(\mathbf{h}_i, \mathbf{h}_p)/\tau}) + \sum_{\mathbf{h}_k \in \mathbf{K}} (e^{\text{sim}(\mathbf{h}_i, \mathbf{h}_k)/\tau})} \quad (9)$$

The intra-sample and inter-sample contrastive learning losses are defined as:

$$\mathcal{L}_{CL} = (1 - \lambda)\mathcal{L}_{ICL} + \lambda\mathcal{L}_{OCL} \quad (10)$$

The cross-entropy loss for each sample is:

$$\mathcal{L}_{CE} = \text{CrossEntropy}(\mathbf{W}\mathbf{h} + \mathbf{b}, y) \quad (11)$$

where $\mathbf{W}$ and $\mathbf{b}$ are trainable parameter matrices, and y is the relation between the actual entity pairs of the sample. The total loss is, therefore:

$$\mathcal{L} = (1 - \alpha)\mathcal{L}_{CL} + \alpha\mathcal{L}_{CE} \quad (12)$$

$\mathcal{L}$ is the total training target, α , λ are hyper-parameters that control the relative weights between various losses.

The overall training objective consists of two parts: the first part is a contrastive loss function of semantic representation, which can learn the feature differences between intra- and inter-samples. The second part of the cross-entropy loss can emphatically distinguish the types of samples. In this way, $\mathcal{L}$ can effectively improve the generalization ability of the model.

IV. EXPERIMENT AND ANALYSIS

We design an experimental scenario using realistic datasets to evaluate the CEE model. In the following, we present the dataset and experiment settings, then conduct an empirical study and analyze the results.

A. DATASETS AND EXPERIMENT DESIGN

In this subsection, we first present the data source and then the distribution of the number of entity pairs with relations in each sentence. Next, we present the distribution of sentence length. Finally, we present the CEE model parameter settings.

The experimental data mainly comes from authoritative open data sources such as the National Medical Products Administration¹ and Medlive.² Pharmaceutical experts manually label ten fine-grained drug entity relations for clinical medication guidance. A total of 5017 high-quality drug entity-relation data are used as training data.

Each entity pair in a sentence (and its relations) forms an entry in the training data. Therefore, multiple data entries can be generated from one sentence if there is more than one entity pair. The distribution of valid entity pair numbers in a sentence in the training set is shown in Figure 5.

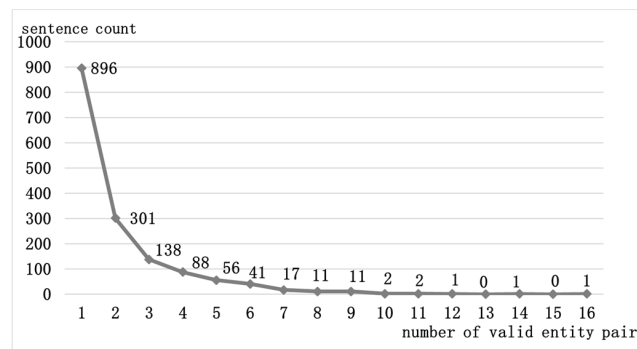


FIGURE 5. Distribution of the number of related entity pairs in sentences.

It can be seen that there are 670 sentences with two or more entity pairs, creating 2284 samples, and the sample generator can create 2284 samples from these 670 sentences, which is far more than the 896 samples generated by sentences with only one entity pair. In an extreme case, one sentence contains up to 16 entity pairs. The density of entity pairs and their relations demonstrates the importance of utilizing the features within and across these entity pairs. Further

¹<http://english.nmpa.gov.cn/>

²<https://www.medlive.cn/>

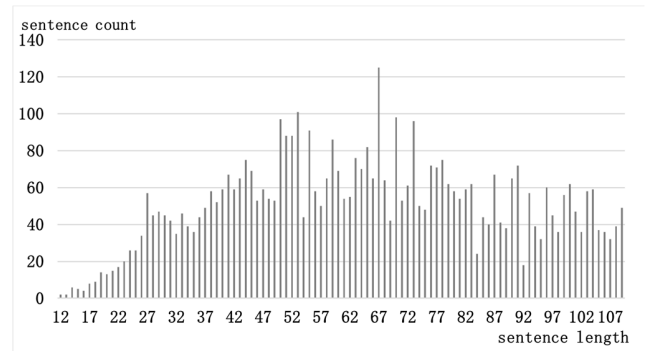


FIGURE 6. Distribution of the sentence length.

analysis of sentence length distribution in the dataset is shown in Figure 6.

It can be seen that the length of sentences in the dataset varies. The shortest sentence has 16 characters. The longest has 109. This leads to very different sentence structures in the negative samples. Combined with the dense entity pairs shown in Figure 5. Since most ($\sim 72\%$) positive samples are generated from the same sentence (has the same length), and most negative ones come from different sentences (varies in length), the model could be biased during training. It tends to believe that the relations between entity pairs in two sentences of the same or similar length are the same. Therefore, the influence of the text length feature on the constructed positive samples needs to be considered.

In addition, we evaluated our model on two public datasets, NYT [39] and WebNLG [40]. The NYT dataset is a corpus of New York Times articles from 1987 to 2007, consisting of 1.18 million sentences and 24 predefined relation types. The WebNLG dataset is a public natural language generation dataset containing 246 predefined relation types.

TABLE 3. CEE parameter settings.

Parameters	Value
Batch size	64
Max sentence length	128
warmup_proportion	0.1
dropout_rate	0.1
Learning_rate	1e-7 - 1e-5
Epochs	25
α	0.5
λ	0.5

The parameter settings of the CEE model are shown in Table 3. The BERT (wwm) model [24] used in CEE consists of 12-layer, 768-hidden, 12-heads, and 110M parameters, where the learning rate adopts the Warmup and Decay [38] learning rate preheat training method.

B. EXPERIMENT AND RESULT ANALYSIS

In this section, we first present the comparative study on CEE and existing models. Then, we discuss the effect of changing

TABLE 4. Comparative experiment results.

Number	Model	Features	Precision	Recall	F1
1	RNN [41]	N/A	0.4284	0.4491	0.4308
2	BERT (wwm) [24]	N/A	0.8116	0.8419	0.8179
3	R-BERT [20]	EN	0.8480	0.8635	0.8467
4	EC-BERT [21]	EN	0.8517	0.8631	0.8483
5	SimCSE(+RD) [10]	IntraCL	0.8416	0.8607	0.8429
6	SCL [31]	InterCL	0.8489	0.8711	0.8471
7	CSCL [34]	InterCL+IntraCL	0.8560	0.8764	0.8562
8	CEE _(+IntraCL+InterCL)	InterCL+IntraCL	0.8592	0.8742	0.8547
9	CEE _(+EN+InterCL)	InterCL+EN	0.8723	0.8860	0.8726
10	CEE _(+EN+IntraCL)	IntraCL+EN	0.8706	0.8804	0.8681
11	CEE _(+ENS)	InterCL+IntraCL+EN	0.8837	0.9015	0.8844

sentence features on the model. Next, we conducted ablation experiments to investigate the effects of the components of the CEE model. Afterward, we discuss the generalization capability of the CEE model in terms of performance under different sample sizes. Finally, we validated the effectiveness of the CEE model in a general domain.

1) COMPARATIVE EXPERIMENTAL RESULTS

The experimental results are shown in Table 4. Precision, Recall, and F1 evaluate the performance of the algorithm model. EN denotes Entity Enhancement, IntraCL denotes intra-sample contrastive learning, and InterCL denotes inter-sample contrastive learning. The sample size in this experiment is set at 50 samples per label.

First, we compare the pre-training model with another deep-learning model. It can be seen that without a sufficient amount of training set. The performance of RNN (#1) is relatively poor. The precision is 42.84%, the recall rate is 44.91%, and it has an F1 score of 43.08%. Adding a pre-training model with BERT (wwm) (#2) can significantly improve the performance, reaching 81.16%, 84.19%, and 81.79% in P, R, and F1, respectively.

Then, we evaluated the impact of data enhancement on model performance. SimCSE(+RD) (#5) refers to using the R-Drop [7] based on BERT (wwm) (#2) combined with SimCSE. SCL (#6) is standard supervised contrastive learning. CEE_(+EN+InterCL) (#9) outperforms SimCSE(+RD) (#5) by 2.34%, 1.93%, and 2.52% in P, R, and F, respectively. CEE_(+EN+IntraCL) (#10) outperforms SCL in P, R, and F by 2.84%, 1.49%, and 2.55%, respectively. This is because CEE_(+EN+InterCL) (#9) and CEE_(+EN+IntraCL) (#10) have added entity enhancements based on SimCSE (+RD) (#5) and SCL (#6), respectively. This allows the model to identify drug-related entities and their sentence roles better. This fully demonstrates the effectiveness of entity enhancement.

Thirdly, we evaluated the impact of the contrastive learning feature on model performance. R-BERT (#3) added the entity pair type annotation into the BERT encoder, and EC-BERT (#4) added entity information and context into

the BERT encoder. However, neither of them uses contrastive learning. CEE_(+EN+InterCL) (#9) has a 2.59% and 2.43% improvement on F1 compared to R-BERT (#3) and EC-BERT (#4), respectively. The model can minimize differences between samples with the same label. This result proves the validity of inter-sample contrastive learning. CEE_(+EN+IntraCL) (#10) outperforms R-BERT (#3) and EC-BERT (#4) on F1 by 2.14% and 1.98%, respectively. The model can minimize the differences between samples with the same semantics. This result also proves the validity of intra-sample contrastive learning.

Meanwhile, we note that combining inter-sample contrastive learning and intra-sample contrastive learning can also improve the performance of the model. CEE_(+IntraCL+InterCL) (#8) shows a 1.18% and 0.76% improvement on F1 compared with SimCSE(+RD) (#5) and SCL, respectively. Compared with CSCL (#7), CEE_(+IntraCL+InterCL) (#8) shows a decrease of 0.53% in recall and an increase of 0.29% in precision. CSCL (#7) adds label information to intra-contrastive learning, which may lead to a more dispersed model distribution in the embedding space, resulting in higher recall and lower precision. These results demonstrate that combining intra-sample and inter-sample contrastive learning can achieve better performance.

Finally, we assemble the entity enhancement with contrastive learning to examine the performance of the CEE model. CEE_(+ENS) (#11) refers to using inter-sample and intra-sample contrastive loss combined with entity enhancement. From the results, it can be seen that the CEE model with both contrastive loss and entity enhancement obtains the best performance. Compared with the model with only an entity enhancement module [20], [21], the model has increased by 3.77% and 3.61% in F1. Compared with the model with only a contrastive learning module [10], [31], [34], the model has increased by 4.15%, 3.73%, and 2.82% in F1. This is because embedding role information enables the model better to identify drug-related entities and their roles in sentences; intra-sample contrastive learning grasps the differences between its features and other samples, making the representation of samples belonging to the same semantic feature in the latent

space more accurate. Inter-sample contrastive learning can minimize the distance between samples with the same label and maximize the distance between samples with different labels. This allows the model to learn the feature differences within and between samples.

2) EXPERIMENTAL RESULTS OF CHANGING SENTENCE FEATURES

In addition, we vary the features of samples generated from the same sentence to evaluate model performance. The results are shown in Table 5.

TABLE 5. CEE performance using different sentence features.

Number	model	F1
1	CEE _(+ENS)	0.8844
2	CEE _(+ENS+Δentity)	0.8801
3	CEE _(+ENS+Δpunctuation)	0.8861
4	CEE _(+ENS+Δword)	0.8892

CEE_(+ENS+Δword) (#4) refers to changing the sentence length of the same sentence through word segmentation and whole-word repetition based on CEE_(+ENS) (#1). CEE_(+ENS+Δpunctuation) (#3) refers to changing sentence length within the same sentence by random punctuation repetition. CEE_(+ENS+Δentity) (#2) refers to replacing the position of an entity in the same sentence. CEE_(+ENS+Δword) (#4) and CEE_(+ENS+Δpunctuation) (#3) achieved experimental results of 0.8892 and 0.8861 on F1, respectively. This is because the sample generator constructs an unbiased and more diversified positive and negative sample, providing an unbiased and more challenging task for the model. In this way, the model can better align the positive samples and diversify the negative samples. It can also be seen that CEE_(+ENS+Δentity) (#2) is 0.43% lower than CEE_(+ENS) (#1) on F1. This is because, although exchanging the positions of entities can build a variety of positive and negative samples, it will lead to changes in the semantics of sentences. As in the example in Figure 2, changing the positions of “atazanavir” and “nelfinavir” will directly lead to disastrous errors.

3) ABLATION STUDIES

We run ablations to understand better the contribution of each component of the CEE model. We first illustrate the impact of contrastive learning. Then, we elaborated on the effect of the entity enhancement. Finally, we demonstrate the effect of word repetition on the model. The results are shown in Table 6.

After removing the Contrastive Learning module (#4), the performance of CEE_(+ENS) (#1) is reduced by 3.53%. It cannot align positive samples and amplify the difference between negative samples. Moreover, after removing inter-sample Contrastive Learning (#2), the performance of CEE_(+ENS) (#1) is reduced by 1.63%, and after removing

TABLE 6. Ablation study results.

Number	model	F1
1	CEE _(+ENS)	0.8844
2	- Inter Contrastive Learning	0.8681
3	- Intra Contrastive Learning	0.8726
4	- Contrastive Learning	0.8491
5	- Entity Enhancement	0.8547
6	+ Word Repetition	0.8892

inter-sample Contrastive Learning (#3), the performance of CEE_(+ENS) (#1) is reduced by 1.18%. This suggests that it is important for the model to capture not only the differences between its characteristics and those of other samples but also to discover the differences between samples with the same label and those with different labels.

After removing the Entity Enhancement module (#5), the f1 of the model drops to 0.8547. It cannot identify drug-related entities and their role in the sentence. The research shows that the entity information in the sample plays an important role in relation extraction.

After adding Word Repetition (#6), the f1 of the model rises to 0.8892. This result suggests that constructing an unbiased, more diverse positive and negative sample without changing the semantics also positively impacts model learning.

4) GENERALIZATION PERFORMANCE EXPERIMENTAL RESULTS

We evaluate the performance on different training datasets to further examine the model’s generalization capability. In particular, experimental comparisons are made on the training sets of 50, 100, 150, 200, 250, and 300 samples per relation. The evaluation index uses the F1. The experimental results are shown in Figure 7.

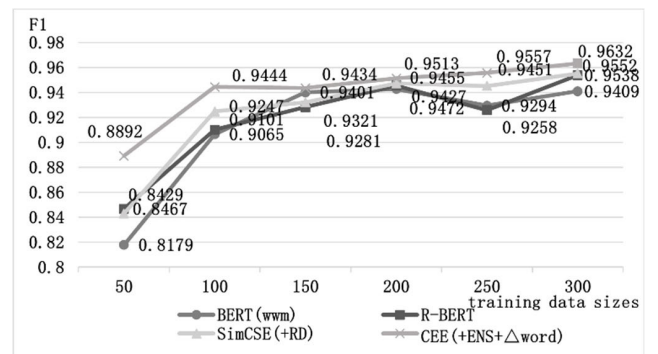


FIGURE 7. Performance of the model with different training data sizes.

It can be seen that when the sample size is 50, the CEE_(+ENS+Δword) model has the largest difference from the experimental results of BERT (wvm), R-BERT, and SimCSE_(+RD) with a maximum difference of 7.13% on F1. It shows that on the training data with a relatively small

TABLE 7. Comparative results in general domain.

Number	Model	NYT (F1)	WebNLG (F1)
full training data			
1	CasRel	0.949	0.940
2	PRCG	0.958	0.945
3	OneRel	0.968	0.968
4	CEE(+ENS+ Δ word)	0.828	0.965
50 training instances of each type			
5	CasRel	0.442	0.832
6	PRCG	0.458	0.796
7	OneRel	0.478	0.837
8	CEE(+ENS+ Δ word)	0.683	0.845

sample size, the CEE(+ENS+ Δ word) model using contrastive loss and entity enhancement has a better generalization capability because it can better capture features and learn their difference. As the sample size increases, the gap between the models gradually decreases. As the sample size increases, it becomes easier to cover more scenarios and learn the differences, reducing this gap.

5) COMPARISON IN THE GENERAL DOMAIN

Table 7 shows the performance results of the CEE model on the NYT and WebNLG datasets compared to other existing models. Our experiments focus only on the relation extraction task and do not consider entity extraction. For a fair comparison, we used a different evaluation protocol, i.e., when a relation is correctly identified, it is considered to be extracted correctly.

When using the complete dataset, our relational extraction model exhibited a significantly lower F1 score on the NYT dataset than other joint extraction models. At the same time, the difference was relatively minor on the WebNLG dataset. This difference can be attributed to the fact that the NYT dataset features approximately one-sixth of the EPO problem (multiple different relations correspond to the same entity pair). In contrast, our previous drug relation extraction task did not involve any EPO issues, which resulted in our model not considering how to address EPO issues.

The few sample experiments in the general domain adopted the same strategy as our main experiments, selecting 50 training instances of each type. With this strategy, the WebNLG few samples experiment consisted of 35% of the complete dataset, while the NYT few samples experiment only contained 1% of the complete dataset. Notably, our model outperforms other joint extraction models, particularly on the NYT dataset, where our model outperforms other joint extraction models by approximately 20% in few-sample scenarios. This is because, in few-sample scenarios, the joint extraction model finds it more challenging to extract the correct relations when only 1% of the NYT training data is available. This further demonstrates the clear advantage of our model in relation to extraction from a few samples.

V. CONCLUSION AND FUTURE WORK

In this paper, a CEE model is proposed by combining entity information enhancement and feature contrastive learning. The CEE model mainly consists of 3 parts: sample generator, entity enhancement, and contrastive learning. The sample generator creates positive and negative samples according to the relation between the same entity pair/sentence and different entity pairs. It controls the characteristics, such as sample length, to create an unbiased training environment. The entity enhancement uses special marks to annotate entity roles based on BERT models. The contrastive learning module employs a combined loss function that captures the intra- and inter-sample feature differences at the same time. Experiments show that this model can achieve a better and more balanced result in drug relation extraction. Compared to other models, when the sample size of each relation is 50, CEE has the best performance, and it gains a 7.13% improvement in F1 compared to BERT (wwm), and a $\sim 3\%$ to $\sim 4\%$ improvement in F1 compared to the SOTA. The generalization experiments with different training sample sizes show that the CEE model has the best generalization performance when the training data size is small. Furthermore, the CEE model is also highly competitive in a general domain.

In our current work, we embed entity information into sentence information, and this embedding needs to be simplified to offer high-quality text vector representation. In future work, we plan to look for more sophisticated methods to enhance entity information, such as combining the type and role information or using existing interaction knowledge as an external feature source. Also, better mechanisms for generating and selecting positive and negative samples for contrastive learning are on the agenda, such as using more syntactic or grammatical features to generate groups of similar positive or more distinctive negative samples. Furthermore, we consider addressing the EPO issue to improve the transferability of our model in a general domain.

REFERENCES

- [1] Z. Zhong and D. Chen, "A frustratingly easy approach for entity and relation extraction," in *Proc. Conf. North Amer. Chapter Assoc. Comput. Linguistics, Human Lang. Technol.*, 2021, pp. 50–61.
- [2] Z. Wei, J. Su, Y. Wang, Y. Tian, and Y. Chang, "A novel cascade binary tagging framework for relational triple extraction," in *Proc. 58th Annu. Meeting Assoc. Comput. Linguistics*, 2020, pp. 1476–1488.
- [3] H. Zheng, R. Wen, X. Chen, Y. Yang, Y. Zhang, Z. Zhang, N. Zhang, B. Qin, X. Ming, and Y. Zheng, "PRGC: Potential relation and global correspondence based joint relational triple extraction," in *Proc. 59th Annu. Meeting Assoc. Comput. Linguistics, 11th Int. Joint Conf. Natural Lang. Process.*, 2021, pp. 6225–6235.
- [4] Y. M. Shang, H. Huang, and X. Mao, "OneRel: Joint entity and relation extraction with one module in one step," in *Proc. AAAI Conf. Artif. Intell.*, 2022, vol. 36, no. 10, pp. 11285–11293.
- [5] F. Gao, L. Zhou, and J. Gu, "Entity pair recognition using semantic enrichment and adversarial training for Chinese drug knowledge extraction," in *Proc. 2nd Int. Symp. Artif. Intell. Med. Sci.*, Oct. 2021, pp. 35–42.
- [6] R. Johnson and T. Zhang, "Deep pyramid convolutional neural networks for text categorization," in *Proc. 55th Annu. Meeting Assoc. Comput. Linguistics*, 2017, pp. 562–570.
- [7] L. Wu, J. Li, Y. Wang, Q. Meng, T. Qin, W. Chen, M. Zhang, and T. Y. Liu, "R-Drop: Regularized dropout for neural networks," in *Proc. Neural Inf. Process. Syst.*, vol. 34, 2021, pp. 10890–10905.

- [8] S. Xu, S. Sun, Z. Zhang, and F. Xu, "Research on multi-feature fusion entity relation extraction based on deep learning," *Int. J. Ad Hoc Ubiquitous Comput.*, vol. 39, nos. 1–2, pp. 93–104, 2022.
- [9] A. Karimi, L. Rossi, and A. Prati, "AEDA: An easier data augmentation technique for text classification," in *Proc. Findings Assoc. Comput. Linguistics*, 2021, pp. 2748–2754.
- [10] T. Gao, X. Yao, and D. Chen, "SimCSE: Simple contrastive learning of sentence embeddings," in *Proc. Conf. Empirical Methods Natural Lang. Process.*, 2021, pp. 6894–6910.
- [11] T. Zhang, Z. Cai, C. Wang, M. Qiu, B. Yang, and X. He, "SMed-BERT: A knowledge-enhanced pre-trained language model with structured semantics for medical text mining," 2021, *arXiv:2108.08983*.
- [12] Y. Yan, R. Li, S. Wang, F. Zhang, W. Wu, and W. Xu, "ConSERT: A contrastive framework for self-supervised sentence representation transfer," in *Proc. 59th Annu. Meeting Assoc. Comput. Linguistics, 11th Int. Joint Conf. Natural Lang. Process.*, 2021, pp. 5065–5075.
- [13] J. Zhang and Z. Lan, "S-SimCSE: Sampled sub-networks for contrastive learning of sentence embedding," 2021, *arXiv:2111.11750*.
- [14] A. Culotta and J. Sorensen, "Dependency tree kernels for relation extraction," in *Proc. 42nd Annu. Meeting Assoc. Comput. Linguistics*, 2004, pp. 423–429.
- [15] R. Bunesu and R. J. Mooney, "Subsequence kernels for relation extraction," in *Proc. NIPS*, Dec. 2005, pp. 171–178.
- [16] D. Zeng, K. Liu, S. Lai, G. Zhou, and J. Zhao, "Relation classification via convolutional deep neural network," in *Proc. 25th Int. Conf. Comput. Linguistics, Tech. Papers*, 2014, pp. 2335–2344.
- [17] M. Peters, M. Neumann, M. Iyyer, M. Gardner, C. Clark, K. Lee, and L. Zettlemoyer, "Deep contextualized word representations," in *Proc. Conf. North Amer. Chapter Assoc. Comput. Linguistics, Human Lang. Technol.*, 2018, pp. 1–30.
- [18] A. Vaswani, N. Shazeer, N. Parmar, J. Uszkoreit, L. Jones, A. N. Gomez, L. Kaiser, and I. Polosukhin, "Attention is all you need," in *Proc. Adv. Neural Inf. Process. Syst.*, 2017, pp. 5998–6008.
- [19] J. Devlin, M.-W. Chang, K. Lee, and K. Toutanova, "BERT: Pretraining of deep bidirectional transformers for language understanding," in *Proc. ACL*, 2019, pp. 4171–4186.
- [20] S. Wu and Y. He, "Enriching pre-trained language model with entity information for relation classification," in *Proc. 28th ACM Int. Conf. Inf. Knowl. Manag.*, Nov. 2019, pp. 2361–2364.
- [21] Y. Wan, L. Sun, P. Zhao, J. Wang, and S. Tu, "Relation classification based on information enhanced BERT," *J. Chin. Inf. Process.*, vol. 35, no. 3, pp. 69–77, 2021.
- [22] S. Wang, B. Zhang, Y. Xu, Y. Wu, and B. Xiao, "RCL: Relation contrastive learning for zero-shot relation extraction," in *Proc. Findings Assoc. Comput. Linguistics*, 2022, pp. 2456–2468.
- [23] Q. Chen, R. Zhang, Y. Zheng, and Y. Mao, "Dual contrastive learning: Text classification via label-aware data augmentation," 2022, *arXiv:2201.08702*.
- [24] Y. Cui, W. Che, T. Liu, B. Qin, and Z. Yang, "Pre-training with whole word masking for Chinese BERT," *IEEE/ACM Trans. Audio, Speech, Language Process.*, vol. 29, pp. 3504–3514, 2021.
- [25] Y. Zhang, X. Liao, L. Chen, H. Kang, Y. Cai, and Q. Wang, "Multi-BERT-wm model based on probabilistic graph strategy for relation extraction," in *Proc. HIS*, 2021, pp. 95–103.
- [26] T. Chen, S. Kornblith, M. Norouzi, and G. Hinton, "A simple framework for contrastive learning of visual representations," in *Proc. ICML*, 2020, pp. 1597–1607.
- [27] X. Chen, H. Fan, R. Girshick, and K. He, "Improved baselines with momentum contrastive learning," 2020, *arXiv:2003.04297*.
- [28] Z. Wu, Y. Xiong, S. X. Yu, and D. Lin, "Unsupervised feature learning via non-parametric instance discrimination," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, Jun. 2018, pp. 3733–3742.
- [29] D. Wang, N. Ding, P. Li, and H. Zheng, "CLINE: Contrastive learning with semantic negative examples for natural language understanding," in *Proc. 59th Annu. Meeting Assoc. Comput. Linguistics, 11th Int. Joint Conf. Natural Lang. Process.*, 2021, pp. 2332–2342.
- [30] P. Khosla, P. Teterwak, C. Wang, A. Sarna, Y. Tian, P. Isola, A. Maschinot, C. Liu, and D. Krishnan, "Supervised contrastive learning," in *Proc. Adv. Neural Inf. Process. Syst.*, vol. 33, 2020, pp. 18661–18673.
- [31] B. Gunel, J. Du, A. Conneau, and V. Stoyanov, "Supervised contrastive learning for pre-trained language model fine-tuning," in *Proc. ICLR*, 2021, pp. 1–15.
- [32] D. Miao, J. Zhang, W. Xie, J. Song, X. Li, L. Jia, and N. Guo, "Simple contrastive representation adversarial learning for NLP tasks," 2021, *arXiv:2111.13301*.
- [33] X. Wu, C. Gao, L. Zang, J. Han, Z. Wang, and S. Hu, "ESimCSE: Enhanced sample building method for contrastive learning of unsupervised sentence embedding," 2021, *arXiv:2109.04380*.
- [34] J. Chi, W. Shand, Y. Yu, K. Chang, H. Zhao, and Y. Tian, "Conditional supervised contrastive learning for fair text classification," in *Proc. Findings Assoc. Comput. Linguistics*, 2022, pp. 2736–2756.
- [35] L. Gong, X. Liu, and Z. Gao, "Drug interaction relationship extraction based on bidirectional GRU and CNN," *J. Shaanxi Normal Univ.*, vol. 48, no. 6, pp. 108–113, 2020.
- [36] L. Huang, J. Lin, X. Li, L. Song, Z. Zheng, and K.-C. Wong, "EGFI: Drug-drug interaction extraction and generation with fusion of enriched entity and sentence information," *Briefings Bioinf.*, vol. 23, no. 1, Jan. 2022, Art. no. bbab451.
- [37] Z. Yuan, Y. Liu, C. Tan, S. Huang, and F. Huang, "Improving biomedical pretrained language models with knowledge," in *Proc. 20th Workshop Biomed. Lang. Process.*, 2021, pp. 180–190.
- [38] K. He, X. Zhang, S. Ren, and J. Sun, "Deep residual learning for image recognition," in *Proc. CVPR*, 2016, pp. 770–778.
- [39] S. Riedel, L. Yao, and A. McCallum, "Modeling relations and their mentions without labeled text," in *Proc. Joint Eur. Conf. Mach. Learn. Knowl. Discovery Databases*, Barcelona, Spain, Sep. 2010, pp. 148–163.
- [40] C. Gardent, A. Shimorina, S. Narayan, and L. Perez-Beltrachini, "Creating training corpora for NLG micro-planners," in *Proc. 55th Annu. Meeting Assoc. Comput. Linguistics*, 2017, pp. 179–188.
- [41] H. Zhang, L. Xiao, Y. Wang, and Y. Jin, "A generalized recurrent neural architecture for text classification with multi-task learning," in *Proc. 26th Int. Joint Conf. Artif. Intell.*, Aug. 2017, pp. 1–7.



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